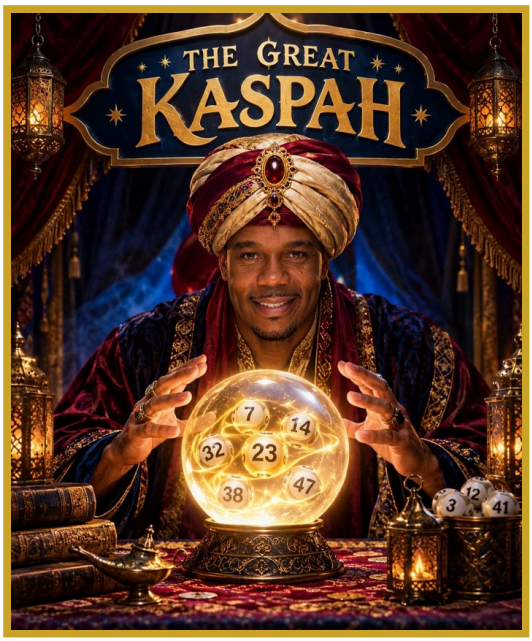
# The Great Kaspah

**Lotto numbers, revealed - when the crystal ball feels like it.**

An entertainment app in which The Great Kaspah, a mystic with a crystal ball, offers a logged-in user a set of lotto numbers for a lotto saved to their profile - together with a flowery sentiment in his voice and a star-rated sense of how strongly he feels about that reading. On some requests he offers nothing at all, because the numbers are not right today. That refusal is part of the product, not an error.

7 14 23 32 38 47

The six balls above are the example set shown on the launch poster. They illustrate the format of a reading only; they are not a recommendation and carry no greater chance of winning than any other valid combination.




**Product:** The Great Kaspah **Hosting:** Vercel **Database:** Neon Postgres
**Status:** Requirements for v1 (MVP) - v2.1 revision, decisions confirmed by the product owner
**Audience:** Product owner, development partner, and testers

**What changed in v2.1:** Round-2 answers are now firm requirements. The logged-in home view is presented in **rows** - one self-contained row per saved lotto (Section 6.4). A **guest mode** lets a visitor receive one reading on a single page after a deliberate theatrical delay that promotes account creation (Section 6.5). Completed readings can be **shared** via a public link back to this site (Section 6.6). Logged-in users may save **personal touches** for Kaspah to weave into readings (Section 6.7). **Date of birth is information only** - never a gate. **V1 is English only.** The starter Kaspah message library is drafted by the team and approved finally by Paul's buddy, the face of Kaspah (D-05). See the changelog in Section 1.4.

# Executive summary

The product has three jobs. First, let a user create an account, log in, and build a profile that records who they are and where they are from. Second, let that user choose one or more lottos - but only from the lottos whose source country matches their profile country - and save those choices to their profile, where they can be changed later. Third, once a profile exists, let the user ask The Great Kaspah for a reading and receive either a valid set of numbers with a flowery sentiment message and a star confidence indicator, or an in-character refusal for that request. The logged-in presentation is in rows - one row per saved lotto, each row holding its own numbers and feeling (Section 6.4). A visitor without an account can also receive a single guest reading through the guest flow in Section 6.5, and a completed reading can be shared through a public share page (Section 6.6).

Version 1 generates numbers randomly. No external data source or prediction API is integrated in v1. Generation sits behind a provider interface so a future API-based provider can replace the random one without changing login, profiles, country filtering, lotto selection, or the reading experience.

The app does not sell tickets, take payment, check draw results, or send notifications in v1. Access is unlimited and free for all users in v1.

> **Read this first - entertainment only.** The Great Kaspah is entertainment. Numbers are generated at random in v1, no reading improves the odds of winning any lottery, and nothing in the app is financial or gambling advice. Section 12 sets the exact disclaimer and responsible-use requirements that must ship with v1.

## How to read this document

- **"Must" is a firm v1 requirement.** Statements marked as decisions in this version were confirmed by the product owner on 14 October 2026 and are not proposals.
- **Numbered requirements (FR, NFR)** are the testable contract for build and acceptance. Section 13 maps acceptance criteria back to them.
- **Pending items are labelled as such** and collected in Section 14.3. Nothing pending blocks the v1 build except where a section says so explicitly. Two items are explicitly proposals, not confirmed figures: the exact guest-delay duration (Section 6.5) and the personal-touch number-inclusion mechanics (D-06).
- **Configuration is not code.** The lotto catalogue, refusal rate, confidence weighting, and message libraries are data or settings, changeable without rewriting application logic.

# 1 The product, the persona, and what changed in v2

## 1.1 Who Kaspah is

The face of the app is the product owner's buddy, presented as The Great Kaspah - a mystic in a gold turban and deep-red robe, crystal ball in hand. The poster supplied for the project sets the visual identity: gold and red mystic styling, a glowing crystal ball, floating lotto balls, and a playful fortune-teller tone. In-app artwork, colours, and copy should follow that identity rather than a generic lottery or finance look.

Kaspah is a character, not a system voice. System messages (for example, a failed login) stay plain and clear, but everything about a reading - the introduction, the flowery sentiment message, the confidence wording, and a refusal - is written and presented as Kaspah speaking. Users should always be able to tell the difference between Kaspah performing and the app informing them.

## 1.2 What the product is - and is not

**It is:**

- A personalised, character-led number-picking experience.
- A profile-based app: your country, your lottos, your readings, your history.
- A repeatable ritual - return whenever you want another reading; v1 places no limit on requests.

**It is not:**

- A ticket seller, bookmaker, or payment service.
- A prediction, tipster, or odds-improvement service.
- A source of financial advice.
- A draw-result checker in v1.

## 1.3 Changelog - v1 to v2

| Area | v1 position (superseded) | v2 firm position |
|---|---|---|
| Market scope | Launch jurisdictions undecided; examples centred on Canada and the US. | **Global.** Country is a required profile field; lottos are filtered by source country (Sections 4, 5). |
| Lotto list | Example seed list, final list undecided, framed as a pre-launch decision. | **Default list ships for now** and will be properly redefined later (pending decision D-01). Maintained by direct database access only; no admin panel in v1 (Sections 4.4, 4.5). |
| Access limits | Open question whether any per-day limit applies. | **Unlimited requests in v1.** No per-day or per-draw cap, and no lockout after a refusal (Sections 2, 6.3). |
| Refusal rate | Proposed default of 15%. | **10%**, stored as a configurable setting (Section 6.3). The earlier figure is superseded and appears nowhere else in this version. |
| Confidence | Stars with a label; flowery wording as flavour. | **Flowery descriptive text is the primary expression**, with stars as visual shorthand. Both are always shown together (Section 6.2). |

## 1.3 Changelog - v1 to v2 (continued)

| Area | v1 position (superseded) | v2 firm position |
|---|---|---|
| Profile fields | Display name and disclaimer acknowledgement only. | **Display name, country, timezone, and date of birth** are all profile fields in v1 (Section 5.2). |
| Authentication | Email and password assumed for v1. | **Confirmed: email + password only.** No social login and no magic link in v1 (Section 5.1). |
| Account deletion | Open question: self-serve or on request. | **Self-serve deletion on the site in v1**, with confirmation and defined data handling (Section 5.6). |
| Money | No payments in v1. | **Free for all users in v1.** Free/paid tiers that could limit requests are a future design only, not built in v1 (Section 2.3). |
| Results checking | Listed as future / undecided. | **Explicitly out of v1.** V1 just gives numbers. Draw-result checking remains future work (Section 2.2). |
| Notifications | Email reminders listed as a v1.1 candidate. | **No notifications of any kind in v1** - no email, push, or draw reminders (Section 2.2). |
| Future algorithm | TODO-01: replace the random provider with an API provider, details TBD. | TODO-01 stands. **No provider has been chosen yet**; the product owner's team is working on it. In the future state the app will request numbers from the algorithm via the API and the algorithm will also offer its understanding - its sentiment and explanation (Section 7.3). |
| Open questions | Q-01 to Q-05 open. | All five are **closed by the answers above** except the genuinely pending items now listed as D-01 to D-05 in Section 14.3. |

Everything not listed here carries over from v1 unchanged: the persona, the provider-interface design, the entertainment-only positioning, the Vercel + Neon stack, and the requirement that every reading - numbers or refusal - is logged through the same pipeline.

## 1.4 Changelog - v2 to v2.1 (round-2 answers)

| Area | v2 position (superseded where stated) | v2.1 firm position |
|---|---|---|
| Logged-in presentation / primary lotto | Ask Kaspah page: user chooses one saved lotto per request; no presentation model specified. Primary-lotto question open. | **Rows presentation.** One self-contained row per saved lotto; each row has its own request, numbers / feeling, or refusal. **There is no single primary lotto.** (Section 6.4, FR-26 to FR-30) |
| Kaspah lines | Starter library structure in Section 6; final approval pending (D-05). | **Approach confirmed:** team drafts a starter library of flowery sentiment and refusal lines; Paul's buddy, the face of Kaspah, gives final approval. D-05 remains the only open item on the lines. |
| Date of birth | Collected as an age-eligibility reference; per-country age wording / gate pending. | **Superseded: DOB is information only.** No gate, no block, no warning, no verification claim, and no use in generation, personalisation, or marketing. Disclaimers are shown to everyone regardless of DOB. (FR-07, Section 12) |
| Language | Not specified. | **English only for v1** - UI, Kaspah messages, guest flow, and share pages. Other languages are future only. (FR-46) |
| Guest access | All reading functions blocked for logged-out visitors (old FR-04). | **Guest mode added.** A visitor can receive one singular guest reading after Kaspah asks country, then lotto, followed by a deliberate theatrical delay promoting account creation. No guest profile, rows, or history. (Section 6.5, FR-31 to FR-37) |

## 1.4 Changelog - v2 to v2.1 (continued)

| Area | v2 position | v2.1 firm position |
|---|---|---|
| Sharing | Social sharing listed as out of scope. | **Share links added.** A completed numbers reading can be shared via a unique, unguessable, revocable public link to a page on this site, ending with: "Want your own numbers? Create an account and log in." Strict privacy rules apply. (Section 6.6, FR-38 to FR-42) |
| Personal touches | Personal data never influenced a reading. | **Personal touches added** for logged-in users: optional lucky numbers, names / words, and significant dates, saved on the profile and woven into readings only. Exact number-inclusion mechanics are a **proposed default pending confirmation (D-06)**. DOB itself is never a personal touch input. (Section 6.7, FR-43 to FR-45) |
| Buddy / persona sign-off | Persona direction described; sign-off status not recorded. | **Confirmed:** Paul's buddy is on board and has loved the ideas; persona and likeness direction is approved in principle. Formal written consent for commercial use should be on file before go-live (housekeeping, not a blocker). No domain name has been provided yet. (Section 14.3) |
| Unchanged from v2 | Global country filtering, multiple lottos per profile, unlimited free access, 10% refusal rate, flowery text first with stars, email + password only, self-serve deletion, no notifications, no result checking in v1, catalogue by direct database access only, and TODO-01 (no API provider chosen yet) all stand exactly as in v2. | |

# 2 Scope: MVP, out of scope, and roadmap

## 2.1 In scope for v1 (MVP)

- Account creation, login, logout, and password recovery, email + password only (FR-01 to FR-04).
- Profile creation and editing: display name, country, timezone, date of birth, and saved lotto selections (FR-05 to FR-13).
- A global, country-filtered lotto catalogue: every lotto carries a source country, and a user is offered only the lottos for their profile country (FR-14 to FR-17).
- A pre-populated default lotto list (Section 4.4), maintained by direct database access only (FR-16, FR-17).
- Requesting a Kaspah reading for a saved lotto, with **no limit on the number of requests** (FR-18).
- Readings that return either valid numbers with a flowery sentiment and star confidence, or a logged refusal at a 10% rate (FR-19 to FR-22).
- Reading history for the signed-in user (FR-23).
- Self-serve account deletion on the site (FR-24, FR-25).

## 2.1 In scope for v1 (MVP) - continued

- **Rows presentation** for the logged-in home / readings view: one self-contained row per saved lotto (FR-26 to FR-30, Section 6.4).
- **Guest mode:** a single guest reading from the landing page - Kaspah asks country, then lotto - shown on a singular result page after a deliberate, configurable theatrical delay that promotes account creation (FR-31 to FR-37, Section 6.5).
- **Sharing:** a unique, revocable share link for a completed numbers reading, opening a public page on this site with a create-account call to action (FR-38 to FR-42, Section 6.6).
- **Personal touches** for logged-in users: optional lucky numbers, names / words, and significant dates saved on the profile and used only in that user's readings (FR-43 to FR-45, Section 6.7).
- **English only:** all user-facing copy, Kaspah messages, the guest flow, and share pages are in English in v1 (FR-46).
- Random generation only, behind a `PredictionProvider` interface (Section 7).
- Entertainment-only disclaimer and responsible-use copy at the points defined in Section 12.

## 2.2 Out of scope for v1

- Selling tickets, linking a purchase to a specific retailer, payments, wallets, subscriptions, or handling money in any form.
- Free/paid tiers or any other mechanism that limits how many readings a user may request. Tiers are a future design consideration only (Section 2.3).
- Any external prediction API, statistical model, draw-history analysis, or "hot number" logic.
- Checking numbers against draw results, match or winnings verification, win notifications, or prize claims. **V1 just gives numbers.**
- **Notifications of any kind** - no email notifications, no push, and no draw reminders. The only permitted email is transactional account email required for password reset and, if implemented, deletion confirmation.
- An admin panel or admin UI of any kind. Catalogue maintenance is by direct database access (Section 4.5).
- Social login, magic-link login, leaderboards, groups, or syndicates. Sharing in v1 means only the site-generated share link in Section 6.6 - there is no posting to external social networks from inside the app in v1.
- **Any language other than English.** No translated UI, messages, guest flow, or share pages in v1, despite the global audience.
- **Any date-of-birth gate.** No user is blocked, warned, or treated differently because of their date of birth in v1 (FR-07).
- **Guest accounts in any form:** no guest rows view, no guest history, no saved guest selections, no guest personal touches, and no way to recover a guest reading after leaving the guest result page - guest readings are not shareable in v1 (Section 6.5).
- Native mobile apps; v1 is a responsive web app hosted on Vercel.
- Multiple profiles per account, family accounts, or sharing a profile. A user who lives in more than one country still has one profile country at a time (Section 4.2).
- Free-text lotto entry anywhere in the product.

## 2.3 Roadmap split

| Phase | Contents |
|---|---|
| **V1 - MVP** | Everything in Section 2.1, including rows presentation, guest mode with deliberate delay, share links, and personal touches. Random provider only. Global catalogue with a default lotto list. Unlimited, free access for logged-in users. English only. No notifications, no result checking, no admin panel, no tiers. |
| **Future - algorithm via API (TODO-01)** | `ApiPredictionProvider` replacing the random provider. In that future state the app requests numbers from the algorithm through the external API, and the algorithm also offers its understanding - its sentiment and explanation for the reading - rather than the app picking a message from a local library. Provider, authentication, contract, cost, and fallback behaviour are all TBD; no provider has been chosen yet. |
| **Future - lotto list redefinition (D-01)** | The default lotto list is properly redefined: which countries, which games per country, and verified rules for each. This is a data exercise, not a rebuild, because the catalogue is already country-keyed data. |
| **Future - free / paid tiers** | Tier design that could limit request volume for free users is a later consideration. It is **not built in v1**. It needs no redesign when it comes, because every request is already logged and countable in `prediction_requests` (Section 8.2); a future limit can be enforced by counting that table. |
| **Future / undecided** | Draw-result checking and match verification, notifications and draw reminders, an admin panel for catalogue maintenance, social or magic-link login, native apps, and **languages other than English**. None is committed by this document, and none may be presented to users as coming in v1. |
| **Future - personal touches confirmation (D-06)** | Confirm or amend the proposed default in Section 6.7 for how a valid lucky number may be included in a generated set. The sentiment-weaving part of personal touches ships in v1 regardless; only the exact number-inclusion mechanics await Paul / buddy confirmation. |

> **Scope discipline.** If a feature is listed in Section 2.2 or in a future row above, it is not a v1 defect when it is missing. Testers should raise it only if v1 copy, screens, or marketing imply that it exists.

# 3 Users and user stories

| ID | User story | Acceptance signal |
|---|---|---|
| US-01 | As a new visitor, I want to create an account and log in, so my profile and readings are saved for me. | I can register, log in on a second device or browser, and see the same profile. |
| US-02 | As a new user, I want to tell the app where I am from during setup, so I am only offered lottos I can actually play. | Country is required before any lotto is shown, and the selection list contains only lottos whose source country matches my profile country. |
| US-03 | As a new user, I want to pick my lottos from that country-filtered list, so Kaspah only reads for games I play. | At least one lotto must be selected before the first reading; selections appear on my profile. |
| US-04 | As a returning user, I want to change my lotto selections and my profile details, so my profile stays current. | Edits save, persist after logout and login, and apply to the next reading request. |
| US-05 | As a user who moves country, I want to change my profile country and understand what happens to my old selections. | I am warned before saving; selections for the old country are removed or deactivated, and my reading history is unchanged (Section 4.2). |
| US-06 | As a player, I want to ask Kaspah for a reading, so I get numbers in the correct format for my lotto. | Returned numbers always match the lotto's count, range, and bonus rules (FR-20). |
| US-07 | As a player, I want to ask as often as I like, so a refusal or a reading I dislike never blocks me. | No request is refused for volume reasons in v1, and after a refusal I can request again immediately. |
| US-08 | As a player, I want to know how strongly Kaspah feels, in his own flowery words, so the experience feels like a character's judgement, not a printout. | Every numbers result shows a prominent flowery descriptive phrase and a star rating together (Section 6.2). |

| ID | User story | Acceptance signal |
|---|---|---|
| US-09 | As a player, I want Kaspah to sometimes decline, so his readings feel considered - without being locked out afterwards. | A refusal shows an in-character message, no numbers and no confidence, and a working "ask again" path with no waiting period (Section 6.3). |
| US-10 | As a player, I want to see my past readings, including refusals, so I can revisit what Kaspah said. | History lists date, lotto, numbers or refusal, flowery sentiment, and stars. |
| US-11 | As a user, I want to delete my account myself on the site, without emailing anyone. | Settings offers deletion with a clear confirmation step; after deletion I cannot log in and my profile, selections, and history are gone (Section 5.6). |
| US-12 | As the product owner, I want the lotto list to be country-keyed data in the database, so I can add a country or a lotto by editing data, not code. | Adding a country and a lotto row by direct database access makes that lotto selectable by users of that country, with no code release (Section 4.5). |
| US-13 | As the product owner, I want every reading request counted, even though v1 has no limits, so a future tier can be designed from real usage. | `prediction_requests` holds one row per request with user, lotto, timestamp, and outcome, queryable by user and period. |

## 3.1 Who the users are

**Primary user:** an adult lottery player, anywhere in the world, who already plays - or is eligible to play - at least one lotto offered for their country, and who enjoys a character-led ritual around picking numbers. They are not looking for an edge, and the product must never suggest they have found one.

**Product owner / operator:** Paul and his team. In v1 the operator is also the catalogue maintainer, working directly in the database (Section 4.5), and the team that drafts Kaspah's starter message library, with Paul's buddy giving final approval (pending decision D-05).

**Not a v1 user:** syndicates or groups sharing one account, and retailers or operators - the app has no partner or B2B surface in v1. The app sells no tickets and applies **no date-of-birth gate** in v1 (FR-07); Section 12 states the entertainment and responsible-use wording shown to everyone.

## 3.2 User stories - round-2 additions (v2.1)

| ID | User story | Acceptance signal |
|---|---|---|
| US-14 | As a logged-in player, I want my home view in rows - one row for each lotto I saved - so I can ask, and see the feeling, for each game in its own place. | One row per saved lotto, in saved order; each row shows its own request action and its own result state, independent of every other row (Section 6.4). |
| US-15 | As a visitor without an account, I want Kaspah himself to ask where I am from and which lotto I want, so I can try one reading without signing up. | Guest flow asks country first, then offers only that country's lottos; one singular result page follows (Section 6.5). |
| US-16 | As a guest, I want to understand why signing up is better, in Kaspah's own flowery way. | During the deliberate delay and again after the result, flowery copy explains that creating an account and logging in gives faster results (FR-35, FR-36). |
| US-17 | As a logged-in player, I want to share a reading Kaspah gave me, so someone else can see it on this site and be invited to get their own. | A share link opens a public page showing the reading and "Want your own numbers? Create an account and log in." - and no private profile data (Section 6.6). |
| US-18 | As a logged-in player, I want to give Kaspah personal touches - my lucky numbers, names, and special dates - so his readings feel woven for me. | Touches save on my profile, appear in my readings only, never on share pages, and are deleted with my account (Section 6.7). |
| US-19 | As a user in any country, I accept that v1 speaks English, so I am not promised a version in my own language yet. | Every v1 surface is English only; no language switcher or translated copy ships (FR-46). |

# 4 Countries and the lotto catalogue

The product is **global**. There is no single launch market, and no country is treated as the default for everyone. Instead, country is a first-class piece of data that links a user to the lottos they may choose from.

## 4.1 The country model

- A **countries** reference table holds every supported country: a stable country code (ISO 3166-1 alpha-2, for example `CA`, `US`, `GB`, `IE`, `DE`), a display name, and an `is_active` flag. A country with no active lottos should not be offered at onboarding, because a user from that country could not complete setup.
- Every **profile** stores exactly one `country_code`, collected at onboarding with the question "Where are you from?" (Section 9.1). It is required: onboarding cannot be completed without it.
- Every **lotto** stores exactly one source `country_code`. Multi-country games are the one nuance: a game such as EuroMillions is played in several countries. In v1 this is modelled by one lotto row per source country that offers it - same game, same rules, different `country_code` - so the filtering rule below stays a simple equality check. If that duplication becomes unmanageable, a lotto-to-country join table is a data-model refinement for the redefinition exercise (D-01), not a v1 requirement.
- **The filtering rule is absolute:** a user is offered only lottos where `lottos.country_code = profiles.country_code` and `lottos.is_active = true`. There is no "show all countries" toggle, no browsing other countries' lottos, and no way to save a lotto from another country in v1.

## 4.2 Changing profile country later

A user may change their profile country at any time from My Profile - people move. Because selections are country-bound, a country change is not a silent edit:

1. Before saving, the app must warn the user, in plain system voice, that changing country will remove or deactivate all current lotto selections that belong to the old country, and that they will need to choose new lottos for the new country before their next reading.
2. The user must explicitly confirm. Cancelling leaves the profile and selections untouched.
3. On confirmation, the old-country selections are removed from `user_lotto_selections` (or marked inactive, if the technical design prefers a soft delete - the user-visible effect must be identical: they no longer appear as selected and cannot be used for a reading).
4. **History is never rewritten.** Past readings keep the lotto name and country as they were at the time of the reading, even if that lotto is no longer selectable by this user.
5. Until the user selects at least one lotto for the new country, the app treats onboarding's lotto step as incomplete and routes them to lotto selection before Ask Kaspah.

## 4.3 Required fields for every lotto

| Field | Meaning and rule |
|---|---|
| `name` | Display name as users know the game, e.g. "Lotto 6/49". |
| `country_code` | Required. Foreign key to `countries`. The source country that offers this lotto; it decides which users ever see it. A lotto row cannot be created or activated without it. |
| `jurisdiction_note` | Optional free note for sub-national detail (province, state, or operator) where it helps the operator keep the list correct. Not user-facing in v1 unless the design chooses to show it. |
| `main_count / main_min / main_max` | How many main numbers are drawn, and the inclusive range they are drawn from. |
| `bonus_count / bonus_min / bonus_max / bonus_from_main_pool` | Bonus number rules, or none. If there is no bonus, counts are 0 and ranges are null. If the bonus is drawn from the same pool, it must not duplicate a main number; if it has its own pool, duplication rules follow that pool. |
| `draw_days` | Days of the week the game is drawn, stored as a list (e.g. Wednesday, Saturday). Used for "next draw" wording. |
| `is_active` | Whether the lotto is currently offered for selection to users of its country. |

## 4.4 The default lotto list - for now

The table on the next page is the **default list that ships in v1**. It is starter seed data that proves the country model across several countries; it is **explicitly not the final list**. The list will be properly redefined later (pending decision D-01). Until then:

- Values below are configurable data, never hard-coded in application code.
- Before go-live, the operator must still sanity-check each row's rules against the game's operator - a wrong range in a default row produces valid-looking but wrong readings.
- Adding or removing countries and games later is a data change under Section 4.5, not a rebuild.

| Country | Lotto (default list) | Main | Bonus | Draw days |
|---|---|---|---|---|
| Canada (CA) | Lotto Max | 7 from 1-50 | 1 bonus from remaining pool | Tue, Fri |
| Canada (CA) | Lotto 6/49 | 6 from 1-49 | 1 bonus from remaining pool | Wed, Sat |
| United States (US) | Powerball | 5 from 1-69 | 1 from 1-26 (separate pool) | Mon, Wed, Sat |
| United States (US) | Mega Millions | 5 from 1-70 | 1 from 1-24 (separate pool) | Tue, Fri |
| United Kingdom (GB) | UK National Lottery Lotto | 6 from 1-59 | 1 bonus from remaining pool | Wed, Sat |
| Ireland (IE) | EuroMillions (Ireland entry) | 5 from 1-50 | 2 from 1-12 (separate pool) | Tue, Fri |
| Germany (DE) | EuroMillions (Germany entry) | 5 from 1-50 | 2 from 1-12 (separate pool) | Tue, Fri |

Default list - to be properly redefined later (D-01). The two EuroMillions rows illustrate the Section 4.1 rule: one row per source country for a multi-country game, so Irish users and German users each see the game under their own country. Draw rules shown are the commonly published formats at the time of writing and must be verified by the operator before go-live; lottery operators change ranges and draw days from time to time.

## 4.5 Catalogue maintenance - direct database access only

**There is no admin panel, admin screen, or in-app editor for the lotto list in v1.** The catalogue - countries, lottos, the refusal-rate setting, and the message libraries - is maintained by **direct access to the Neon database** by the product owner or a person the product owner explicitly authorises. Because there is no UI to catch mistakes, the following safeguards are requirements, not advice:

- **Named people only.** Database write access for catalogue tables is limited to named individuals. Credentials are never shared in chat, email, or source code, and are held in the team's password manager / Neon project roles.
- **Validate before it goes live.** A new or edited lotto is inserted with `is_active = false` first. Before activation, the editor checks: country code exists in `countries`; counts and ranges are consistent (count does not exceed pool size; min is below max); bonus rules match the game's real format; draw days are valid. Only then is `is_active` set to true.
- **Test the effect.** After activation, the editor confirms - with a test account whose profile country matches - that the lotto appears in selection and that a test reading validates against its rules.

- **Never delete a lotto that has history.** Retire it with `is_active = false` instead (FR-17). Deleting a row that selections or predictions point to would corrupt users' history. The same applies to countries: a country with profiles or lottos is deactivated, not deleted.
- **Back up before bulk changes.** Before a batch edit (for example the future list redefinition, D-01), take a Neon backup or branch, or export the affected tables, so a bad edit can be rolled back.
- **Keep a change note.** Every catalogue change is recorded in a simple operator change log (date, person, rows changed, reason) kept outside the app. The database is the system of record for the data; the log is the record of who changed it and why - because direct access leaves no in-app audit trail of its own.
- **Production is not the test bed.** Where possible, make and verify the change first on a Neon branch or the preview environment's database, then apply the same change to production.

> **Why this section is strict:** a typo made directly in the database reaches users immediately, with no review screen in between. A wrong country code hides a lotto from everyone who should see it; a wrong range produces invalid readings at scale. The checks above are the control that replaces the missing admin UI.

### 4.6 Retiring lottos and countries

Retiring a lotto (`is_active = false`) removes it from new selection for its country, keeps existing selections readable in the profile (shown as no longer available, and not usable for a new reading), and keeps past predictions displayed with the lotto's name as it was. Retiring a country stops it being offered at onboarding; existing profiles in that country keep working with their history, and the product owner decides separately whether their active lottos remain readable.

# 5 Functional requirements

Requirements are numbered for build and test traceability. "Must" is required for v1 acceptance; "should" is expected unless a documented decision removes it.

## 5.1 Accounts and authentication

| ID | Requirement |
|---|---|
| FR-01 | The system must allow a visitor to register with an email address and password, with the password stored only as a secure hash - never in plain text. |
| FR-02 | The system must allow a registered user to log in and log out, and must keep them logged in across a normal session on the same device. |
| FR-03 | The system must provide a password-reset flow by email for a user who cannot log in. This transactional email is account plumbing, not a notification feature (Section 2.2). |
| FR-04 | The system must block all profile, selection, logged-in reading, and history functions for logged-out visitors, and redirect them to login. **Guest mode (FR-31 to FR-37) is the single exception:** a logged-out visitor may complete the guest flow and view its singular result page, and anyone may view a public share page (FR-38). Neither exception exposes any logged-in function or any other user's data. |
| FR-04a | **Email + password is the only authentication method in v1.** The system must not offer social login, magic-link login, or phone login in v1, and no login screen copy may imply that it does. |

## 5.2 Profile

| ID | Requirement |
|---|---|
| FR-05 | On first login the system must take the user through profile creation (onboarding) before their first reading, in this order: profile details, country, disclaimer acknowledgement, then lotto selection (Section 9.1). |
| FR-06 | The profile must record: **display name** (required), **country** (required, from the `countries` list), **timezone** (required - pre-filled from the device where possible and editable, used to date readings and "today / tomorrow" wording in the user's own time), and **date of birth** (required - see FR-07). |
| FR-07 | Date of birth is **profile information only**. It must **not** act as an age gate, must **not** block, warn, restrict, or treat differently any user, must **not** be presented anywhere as age verification, and must **not** be used in number generation, personalisation (including personal touches, Section 6.7), targeting, or marketing. Entertainment-only and responsible-play disclaimers are shown to **every** user regardless of date of birth (Section 12). DOB must be stored securely, shown only to the account holder in their own profile, and deleted with the account (FR-25). Any v2 wording that gave DOB an eligibility purpose is superseded by this requirement. |
| FR-08 | All profile fields in FR-06 must be editable later from My Profile, except that changing country triggers the warning and selection consequences in Section 4.2 (FR-13). |

## 5.3 Lotto selection

| ID | Requirement |
|---|---|
| FR-09 | Lotto selection must present **only active lottos whose source country equals the user's profile country** (Section 4.1). There must be no free-text lotto entry and no way to view or select another country's lottos. |
| FR-10 | The user must select at least one lotto to complete onboarding, and **may select more than one**. There is no upper limit on selections within the user's country subset in v1. |
| FR-11 | Selected lottos must be persisted to the user's profile ( `user_lotto_selections` ) and must be editable at any later time from the profile screen. Saved changes apply to subsequent reading requests and survive logout and login. |
| FR-12 | If the user's country subset contains no active lottos, the app must say so plainly, explain that readings are not available yet for their country, and let them complete the rest of their profile. It must not offer another country's lottos as a workaround. |
| FR-13 | Changing profile country must follow Section 4.2 exactly: advance warning, explicit confirmation, removal or deactivation of old-country selections, history untouched, and lotto re-selection required before the next reading. |

## 5.4 Lotto catalogue administration

| ID | Requirement |
|---|---|
| FR-14 | Every catalogue entry must carry all fields in Section 4.3, including a valid `country_code` . An entry missing a required rule, or pointing at a non-existent or inactive country, must not be set active. |
| FR-15 | The catalogue must be loaded and changed as seed / reference data in the database, not hard-coded in page or component code. |
| FR-16 | Catalogue changes in v1 are made **only by direct database access** by authorised, named individuals, following the validation, testing, backup, and change-log safeguards in Section 4.5. The system must not include an admin panel or in-app catalogue editor in v1. |
| FR-17 | Retiring a lotto ( `is_active = false` ) must remove it from new selection, keep existing selections readable, and keep past predictions displayed with the lotto's name as it was. Deleting a lotto or country that has selections, profiles, or predictions is prohibited. |

## 5.5 Readings and history

| ID | Requirement |
|---|---|
| FR-18 | A logged-in user with a complete profile must be able to request a reading for any lotto saved to their profile. **Requests are unlimited in v1:** the system must not impose a per-day, per-draw, or lifetime cap, must not charge for requests, and must not block a request because of how many requests the user has already made. |
| FR-19 | A refusal must never act as a lockout. After a refusal the user may request again immediately, for the same lotto or another saved lotto (Section 6.3). |
| FR-20 | A numbers result must conform exactly to the selected lotto's rules: correct count, unique within each pool, within range, bonus rules honoured, and presented sorted in ascending order (Section 6.1). |
| FR-21 | Every numbers result must include a sentiment message in Kaspah's voice whose **primary expression is a flowery descriptive phrase**, together with a star confidence indicator on the scale in Section 6.2. Neither may be shown without the other. |
| FR-22 | The system must sometimes return a refusal instead of numbers, decided as defined in Section 6.3 at a configured rate of **10%**, with an in-character message and guidance to try again. |
| FR-23 | Every request - numbers or refusal - must be logged as a prediction request with its outcome (Section 8.2). The user must be able to view their own reading history, newest first, and must never see another user's history. |
| FR-24a | The entertainment-only disclaimer must appear at onboarding, on the reading screen, and with every reading result (Section 12). |

## 5.6 Self-serve account deletion

| ID | Requirement |
|---|---|
| FR-24 | A logged-in user must be able to delete their own account **on the site, without contacting anyone**: Settings / My Profile, then Delete Account. The flow must (1) explain in plain language what will be deleted and that deletion is **irreversible**; (2) require an explicit confirmation step that cannot be triggered by a single accidental tap (for example, re-entering the password or typing a confirmation word - the exact mechanism is a design choice, but a deliberate second action is mandatory); and (3) complete the deletion and log the user out everywhere. |
| FR-25 | On deletion, the user's account row, profile (including date of birth and timezone), lotto selections, prediction requests, and predictions / reading history are **deleted with the account** - history goes with it, and the user must be told so before confirming. Where a record must be retained for a legal or security reason, it must be anonymised so it no longer identifies the user, and the deletion screen must say so in general terms. After deletion, the same email address may register again as a brand-new account with no recovered data. |

## 5.7 Rows presentation - logged-in home / readings view

| ID | Requirement |
|---|---|
| FR-26 | The logged-in home / readings view must be presented as a **list of rows, one row per lotto saved on the user's profile**. There is no single primary or default lotto that Kaspah uses automatically; the row is the unit of request and result. If the user has saved one lotto, there is one row; if they have saved four, there are four rows. |
| FR-27 | Each row must be **self-contained** and must show: the lotto name and its source country; its draw days; its own request action ("Ask Kaspah" for this lotto); and its own result state for the most recent request made from that row in this view - either (a) its numbers, with bonus numbers separated where applicable, together with its stars and flowery feeling, (b) its own "no numbers today" refusal state with its in-character refusal message, or (c) a not-yet-requested state making clear no reading has been asked for in this row yet. |
| FR-28 | Rows are **independent**. Requesting a reading, receiving numbers, or receiving a refusal in one row must not change, clear, replace, or block any other row. The 10% refusal roll is made **per request, per lotto** (Section 6.3). |
| FR-29 | Rows must appear in the **order the user saved the lottos** (earliest saved first, by `user_lotto_selections.selected_at`), and that order must be stable across logout and login. Removing a lotto removes its row; saving a new lotto appends a new row at the end. |
| FR-30 | No row may show a lotto from another country. The country rule in FR-09 applies to rows exactly as to selection: every row's lotto must satisfy `lottos.country_code = profiles.country_code` at the time it was saved, and a country change clears rows with the selections (FR-13). |

## 5.8 Guest mode

| ID | Requirement |
|---|---|
| FR-31 | A visitor without an account must be able to start a guest reading from the landing page, without registering or logging in. The guest flow is presented by **Kaspah himself, in flowery in-character language**, and asks two things **in order**: (a) "Where are you from?" - a country chosen from the supported countries list; then (b) which lotto they want numbers for, offered **only** from the subset for that chosen country, using the same filtering rule as FR-09. No lotto is shown before a guest country is chosen. |
| FR-32 | Guest country and lotto choices are **request-scoped only**. They must not be stored as a profile, must not create an account or a profile row, and must not be remembered as selections after the guest result page is left, except in the anonymous guest log in FR-37. |

| ID | Requirement (guest mode, continued) |
|---|---|
| FR-33 | The guest receives a **singular result page: one lotto, one outcome** - either numbers with bonus where applicable, stars, and flowery feeling, or a refusal. The same lotto-rule validation (FR-20) and the **same 10% refusal rate** apply to guest requests. The entertainment disclaimer appears on the guest result page. |
| FR-34 | **Deliberate delay.** The guest result must take noticeable time to display, through the theatrical reviewing sequence specified in Section 6.5, driven by the configurable setting `app_settings.guest_delay`. The exact duration is a **tunable proposal** in Section 6.5, not a confirmed figure. Logged-in readings must **not** be subject to this artificial delay. |
| FR-35 | During the guest wait, the page must carry flowery in-character copy to the effect that **Kaspah is reviewing / consulting the crystal ball**, together with the message that **for faster results, the visitor should create an account and log in**. |
| FR-36 | After the guest result is shown, the page must prompt the guest **again**, in flowery language, to create an account for faster processing later, with a clear create-account / log-in call to action. |
| FR-37 | Guests receive **no rows view, no history, no saved selections, and no personal touches** in v1. Guest requests may be logged anonymously (country, lotto, timestamp, outcome - no personal data) for counting and abuse protection. Standard infrastructure rate-limiting may protect the guest endpoint; it is a security control, **not** a product usage cap, and must not be presented to guests as a quota. |

## 5.9 Sharing a reading

| ID | Requirement |
|---|---|
| FR-38 | A logged-in owner of a **completed numbers reading** must be able to generate a **unique, unguessable share link** for that reading (Section 6.6). Refusals have no numbers to share and are not shareable in v1. A share link opens a **public page on this site** - no login is required to view it. |
| FR-39 | The public share page must show only: Kaspah branding; the lotto name and its source country; the numbers (and bonus, separated); the stars and flowery feeling; the reading date; a statement that **the Great Kaspah provided these numbers**; the entertainment disclaimer; and the call to action, in substance: **"Want your own numbers? Create an account and log in."** |
| FR-40 | **Privacy - firm.** A share page must **never** expose the owner's email, date of birth, timezone, personal touches, display name (not shown by default), account details, other readings, or selections. Nothing beyond the FR-39 list may be added to a share page without a new privacy decision. |
| FR-41 | The owner must be able to **revoke** a share link. A revoked link, and any share link belonging to a **deleted account**, must stop working and show a plain "this reading is no longer available" page with no reading data. |
| FR-42 | Share links must not be enumerable: tokens are long, random, and unguessable, a link reveals nothing about the owner's account identifier, and guessing or altering a token must never return another reading. |

## 5.10 Personal touches

| ID | Requirement |
|---|---|
| FR-43 | A logged-in user may **optionally** save personal touches on their profile, in a dedicated Personal Touches area separate from the core profile fields. V1 touch types are: **lucky numbers** (a small list of integers the user supplies); **names / words** they would like Kaspah to weave in (free text, short); and **significant dates** (dates with an optional short label). All touches are optional - a user with no touches receives ordinary readings with no penalty or prompt loop. |
| FR-44 | Personal touches are editable at any time, are stored on the profile, are used for **no purpose other than that user's readings**, are never shown on share pages or to any other user, and are **deleted with the account** (FR-25). Date of birth is **not** a personal touch and must never be fed into this feature, even if the user also enters the same date as a significant date themselves. |
| FR-45 | In a reading, personal touches are used as specified in Section 6.7: sentiment text weaves in names / words (and may reference a significant date) in Kaspah's flowery voice, and the lucky-number inclusion follows the **proposed default in Section 6.7, pending confirmation as D-06**. No touch may be presented as improving the chance of winning, and the disclaimer still appears with every result. |

## 5.11 Language

| ID | Requirement |
|---|---|
| FR-46 | **English only for v1.** All user interface copy, system messages, Kaspah sentiment and refusal messages, the guest flow and its delay copy, share pages, and transactional account email must be in English in v1. The system must not offer a language switcher, auto-translation, or partially translated surfaces in v1. Other languages are a future consideration only (Section 2.3). |

> **Traceability note.** FR-26 to FR-46 are the v2.1 additions. They do not change FR-01 to FR-25, except for the guest / share exception now stated in FR-04 and the superseding date-of-birth wording in FR-07.

# 6 The Kaspah reading

## 6.1 What a numbers result contains

A successful reading shows, on one result view: (1) the lotto's name; (2) the main numbers as balls, sorted ascending; (3) any bonus numbers, visually separated from the main numbers and labelled as bonus; (4) Kaspah's sentiment message, led by its flowery descriptive phrase; (5) the star confidence indicator; (6) the date of the reading in the user's profile timezone and, where draw days are known, the next draw day for that lotto; and (7) the short entertainment-only disclaimer.

Validation is a hard rule, not a display hope. Before a result is saved or shown, the system must check it against the lotto's stored rules - count, uniqueness, range, bonus pool, and sorting. A result that fails validation must never be shown to the user; the request is logged as a system error and the user is asked, in plain system voice, to try again.

> **Example reading (illustrative copy).** Lotto 6/49 - **7 14 23 32 38 47**. Kaspah says: *"The ball glows warm as a lantern in the desert night - these numbers stepped forward, and I did not chase them."* Confidence: **Strong feeling** - 4 of 5 stars. Entertainment only - every valid combination has the same chance of being drawn.

## 6.2 Sentiment and confidence - flowery text first, stars second

The sentiment is a short, in-character passage from Kaspah reacting to the reading. **Its primary expression is a flowery, descriptive phrase** - the kind of line a mystic would actually say, full of mist, moonlight, and glowing crystal. The star rating is the visual shorthand for the same feeling: a quick-glance summary that sits beside the words, never instead of them.

- Every numbers result must show **both together**: the flowery phrase, prominent and readable as the headline of the sentiment, and the stars beneath or beside it.
- V1 selects the sentiment from a curated message library keyed to the confidence level, so the words and the stars always agree. The library is content, editable without changing generation logic. A starter library is drafted by the team and the final library is approved by Paul's buddy, the face of Kaspah (pending decision D-05) and needs enough lines per level that regular users do not see constant repetition.
- Neither the phrase nor the stars is a probability, a statistical estimate, or a claim that one set of numbers is more likely to win than another. No copy, label, tooltip, or help text may present either as one. A line to that effect must appear near the indicator on first use, and the disclaimer appears with every result (FR-24a).

| Level | Flowery phrase (primary) | Example sentiment flavour |
|---|---|---|
| 1 of 5 stars | **Faint whisper through the mist** | "The mist barely parts tonight... but something small and patient is there." |
| 2 of 5 stars | **A gentle glow in the glass** | "A soft light, like a candle behind silk. Kaspah nods, slowly." |
| 3 of 5 stars | **Clear skies over the crystal** | "The ball is clear tonight, the stars sit still, and these numbers rest well." |
| 4 of 5 stars | **The ball glows warm as a lantern** | "Ahh, warmth rises in the glass. These numbers stepped forward on their own." |
| 5 of 5 stars | **The ball blazes like a captured sun** | "Rare and bright! Kaspah has not felt the glass burn like this in many moons." |

In v1 the confidence level is generated alongside the numbers by the random provider (Section 7), weighted so higher levels are less frequent and level 5 is the rarest. Whatever level is shown, the underlying chance of winning is unchanged.

### 6.3 "No numbers today" - the refusal

Sometimes Kaspah looks into the ball and declines: the vibe is not right, the numbers are not there, and forcing a set would be false. The refusal view shows no numbers and no confidence indicator - neither flowery phrase nor stars. It shows (1) an in-character refusal message, varied from a refusal message library so it does not repeat identically; (2) clear guidance that the user may try again - for example, "The mist has not settled. Ask Kaspah again when you are ready"; and (3) the disclaimer.

**How refusal is decided in v1.** The random provider first makes a refusal draw: each reading request has a **10% probability of refusal**, stored as a setting (not hard-coded) and tunable without a code release. If the refusal draw triggers, the provider returns a refusal outcome and no numbers are generated. If it does not trigger, numbers, confidence, and sentiment are generated.

**A refusal is per-request, never a lockout.** Because access is unlimited in v1 (FR-18), a user who receives a refusal may immediately request again - the same lotto or another saved lotto - and each new request makes its own independent 10% refusal draw. There is no waiting period, no daily cap, and no penalty counter in v1. Refusal guidance copy must reflect this: it may stay in character about the mist, but it must never tell the user they are barred from asking again today.

**Logging.** A refusal is a completed reading, not a failed request. It must be logged in exactly the same pipeline as a numbers result (FR-23): a `prediction_requests` row and a `predictions` row recording the refusal outcome, the refusal message shown, the lotto, and the timestamp, with number and confidence fields empty. Refusals appear in the user's history as refusals.

## 6.4 Rows presentation - the logged-in home view in detail

The rows view replaces any idea of a single chosen lotto on the home screen. When a logged-in user with a complete profile opens the app, they see their lottos as a vertical list of rows, in saved order (FR-29). An illustrative layout:

| Row | What the row shows |
|---|---|
| **Lotto 6/49** - Canada<br>Draws: Wed, Sat | **Numbers:** 7 14 23 32 38 47 Bonus: 19<br>**Feeling:** "The ball glows warm as a lantern" - 4 of 5 stars, with Kaspah's flowery sentiment line.<br>[ Ask Kaspah again ] [ Share this reading ] |
| **Lotto Max** - Canada<br>Draws: Tue, Fri | **No numbers today.** "The mist will not settle over this game tonight, dear seeker..." - refusal state for this row only, with its own ask-again action.<br>[ Ask Kaspah again ] |
| **Daily Grand** - Canada (example row) | **Not yet requested.** "Kaspah has not yet gazed upon this game for you."<br>[ Ask Kaspah ] |

Illustrative only: lotto names, numbers, and copy are examples. The point is the structure - each row carries its own lotto identity, action, and latest result state.

- Requesting from a row updates **only that row's** result state; every other row keeps exactly what it was showing (FR-28).
- A row's latest result also appears in History (FR-23); the row is a live view, history is the permanent record.
- On a small screen, rows stack in the same saved order; no row is hidden behind a tab for another lotto.
- If the user saves no lotto (for example after a country change), there are no rows: the view routes them to lotto selection, exactly as in FR-13.

# 6.5 Guest mode - the singular reading, and the deliberate delay

## The guest sequence, step by step

1. On the landing page, a visitor chooses the guest entry point - in Kaspah's voice, for example: *"No account? No matter. Step closer, traveller, and let the Great Kaspah gaze for you - once, as a guest."*
2. **Kaspah asks: where are you from?** The guest picks a country from the supported countries list. Nothing about lottos is shown yet.
3. **Kaspah asks: which lotto do you want numbers for?** Only the active lottos for the chosen country are offered - the same subset rule as for profiles (FR-31). If that country has no active lottos, the guest is told so in character, plainly, and offered a chance to choose a different country; they are never offered another country's lotto as a substitute.
4. The guest confirms, and the **reviewing sequence** begins (below). No account, profile, or selection is created.
5. A **singular result page** appears: one lotto, one outcome - numbers with stars and flowery feeling, or a refusal - plus the disclaimer, and then the post-result account prompt (FR-36).

## The deliberate delay - a theatrical reviewing sequence

The delay is a **designed product feature, not a performance defect**. While Kaspah "reviews", the page shows a crystal-ball visual with a series of flowery progress lines in Kaspah's voice, appearing in sequence - for example:

- *"Hush now... the Great Kaspah turns to the crystal..."*
- *"The mists swirl... shapes move behind the glass..."*
- *"Patience, traveller. The numbers do not hurry for anyone..."*

Throughout the wait, a clear line in the same flowery register carries the commercial message (FR-35), in substance: *Kaspah is reviewing - and know this: those who create an account and log in are seen far faster. The ball favours its friends.* The exact lines are part of the starter library for buddy approval (D-05).

### 6.5 Guest delay - duration, configuration, and limits (continued)

| Item | Specification |
|---|---|
| Setting | `app_settings.guest_delay` - stored as a duration in seconds, read at request time, changeable without a code release under the Section 4.5 change-note discipline. |
| **Proposed default (tunable proposal - not a confirmed figure)** | **Propose 12 seconds, tunable in a range of about 8 to 20 seconds.** Long enough to feel theatrical and to make the logged-in speed advantage real; short enough that a guest does not abandon the page. Paul / the product owner should confirm or adjust this value before go-live; acceptance testing verifies the delay is noticeable and respects the configured value, not that it equals 12 seconds exactly. |
| Progress behaviour | Progress lines rotate during the wait so the page never looks frozen. The result must not be revealed early by refreshing into it, and the sequence must be honest theatre: no fake "searching databases" or similar claims about real analysis. |
| Logged-in contrast | Logged-in readings are **never** given this artificial delay. They return under the normal performance requirement (NFR-03), with only a brief, skippable reveal animation. That contrast is the stated benefit of an account, in the guest copy and nowhere contradicted. |
| After the result | A second, flowery account prompt appears below the guest result (FR-36), in substance: *"See how the ball rewards patience? Create an account, log in, and next time Kaspah will greet you by name - and far, far faster."* With a clear [ Create an account ] and [ Log in ] action. |
| No guest leftovers | No rows, no history, no saved selections, no personal touches, no share link for the guest reading, and no way to re-open the same guest result after leaving the page. A guest who wants any of those must create an account. |

## 6.6 Sharing a reading - the public page in detail

From a row or from History, the owner of a completed numbers reading chooses Share and the site generates the link (FR-38). The owner can copy and send that link by any means they choose - the site itself does not post it anywhere. Anyone opening it sees the share page described in FR-39, headed by Kaspah branding and a statement such as: *"The Great Kaspah provided these numbers"*, followed by the lotto, country, numbers, stars and flowery feeling, and the reading date. The page closes with the entertainment disclaimer and the call to action, in substance: **"Want your own numbers? Create an account and log in."** The owner manages links from the reading in History: view / copy the link, or revoke it (FR-41). Revocation is immediate.

## 6.7 Personal touches - how Kaspah weaves them in

Personal touches make a reading feel personal without pretending to make it luckier. They are set in My Profile, under Personal Touches (FR-43), with plain help text: what each touch is, that it is optional, that it is used only in the user's own readings, and that no touch changes the odds.

### Firm in v1 - the feeling

- **Names / words:** the sentiment and feeling text weaves the user's saved names or words in naturally and addresses the user warmly in Kaspah's flowery voice - for example, a line that wishes the named person well, or rolls a favourite word into the imagery. Woven text is selected / assembled from the approved message library patterns (D-05), never generated as a claim about that person's luck or future.
- **Significant dates:** sentiment copy may warmly acknowledge a saved significant date when it is near (for example an anniversary wish in the feeling text). The date itself is never converted into numbers in v1, and DOB is never used for this (FR-44).
- **Refusals:** refusal copy may also reference a personal touch in character - for example, that the ball is quiet "even for [name]" tonight - varied from the refusal library, with the same ask-again guidance as any refusal.

> **PROPOSED DEFAULT - needs Paul / buddy confirmation (open decision D-06).** For numbers: if the user has saved lucky numbers, and at least one of them is valid for the chosen lotto (inside the main range, and not duplicating another drawn main number), the random provider **may include exactly one** randomly chosen valid lucky number in the generated main set; every other number is drawn fully at random as in Section 7.1. If no lucky number is valid for that lotto, none is included and nothing is said about it. When one is included, the result copy discloses it in character - the user's lucky number *"stepped forward"* - together with the standing clarification that a user-supplied number confers **no advantage** and every valid combination has the same chance. Bonus numbers are never taken from lucky numbers under this proposal. Until D-06 is confirmed, developers must build the inclusion behind a configuration flag that can be turned off without changing the sentiment-weaving behaviour above.

Worked example under the proposal: user saves lucky numbers 7 and 99, and asks in their Lotto 6/49 row (main range 1-49). 99 is invalid and is ignored; 7 is valid and may be the one included number, with the other five main numbers fully random - or, on another request, no lucky number is included at all, because inclusion itself is a random choice, not a guarantee.

# 7 The algorithm: random in v1, replaceable by design

## 7.1 V1 generation - random only

Version 1 contains **no predictive model**. The v1 provider, `RandomPredictionProvider`, works as follows for a validated reading request:

1. Make the refusal draw described in Section 6.3 (10%). On a refusal, return the refusal outcome.
2. Otherwise, draw the required count of unique main numbers uniformly at random from the lotto's stored main range.
3. Draw any bonus numbers uniformly at random according to the lotto's stored bonus rules, honouring same-pool exclusion where it applies.
4. Draw a confidence level (1-5, weighted so higher levels are less frequent, level 5 rarest) and select a matching flowery sentiment message from the library for that level.
5. Return numbers sorted ascending, with bonus numbers separate, for validation under FR-20 before saving and display.

No draw history, frequency analysis, numerology, astrology, or external data influences v1 generation. The user's country, timezone, and date of birth never influence generation - country only filters which lottos are offered, and timezone only affects how dates are shown. The **sole** exception is the user's own explicitly saved **personal touches** (Section 6.7): names / words and significant dates shape only the sentiment text, and lucky numbers shape the numbers only under the proposed default pending D-06. Any other copy or screen that suggests personal data improves a reading must not ship.

## 7.2 The PredictionProvider interface

All generation sits behind one abstraction so the source of a reading can change without changing the rest of the product. At BRD level the contract is:

```
PredictionProvider.requestReading(lotto, userContext) ->  // userContext carries personal
touches for logged-in users only; empty for guests; never carries date of birth
{ outcome: "numbers" | "refusal",
  mainNumbers?: int[], bonusNumbers?: int[],
  confidenceLevel?: 1..5,
  sentimentMessage?: string,
  refusalMessage?: string,
  providerName: string, providerVersion: string }
```

Rules for the abstraction: the reading flow, validation, logging, and history depend only on this contract, never on a concrete provider; the active provider is chosen by configuration, not by branching throughout the code; every saved prediction records which provider and version produced it, so v1 random readings remain identifiable after any future switch; and a provider failure must surface as a plain system error, never as a Kaspah refusal - a refusal is a deliberate outcome, not an error state.

## 7.3 TODO register and the future algorithm

| ID | Item |
|---|---|
| TODO-01 | **Replace `RandomPredictionProvider` with `ApiPredictionProvider`** connecting to an external API. **No provider has been chosen yet** - the product owner's team is working on it. Provider, authentication method, request / response contract, cost, rate limits, and fallback behaviour are all TBD. No API is integrated, stubbed against a real endpoint, or promised for v1. Work starts only after a separate decision records the provider and contract. |
| TODO-01a | **Future-state behaviour:** when TODO-01 is built, the app will request the numbers from the algorithm through the API, and **the algorithm will also offer its understanding** - its sentiment and explanation for the reading - in place of (or alongside) the local message library. The provider contract in Section 7.2 already carries `sentimentMessage`, so this needs no change to the reading flow; the technical design for TODO-01 must confirm how algorithm-supplied text is presented in Kaspah's voice and reviewed for the entertainment-only rules in Section 12, because an external source must not be presented as improving a user's odds unless that claim can be substantiated - which lottery randomness does not support. |
| TODO-02 | Confirm the starter sentiment, refusal, guest-reviewing, and post-guest-result message libraries (drafted by the team; final approval: Paul's buddy, the face of Kaspah) - enough flowery messages per confidence level, and enough refusal lines, that regular users do not see constant repetition. Tracked as pending decision D-05. |
| TODO-03 | Keep the refusal probability (10%) and confidence weighting as configuration values, reviewed after launch using logged request data - not hard-coded, and not changed casually by direct database edit without a change note (Section 4.5). |
| TODO-04 | Design free / paid tiers that could limit request volume - **future only, not built in v1**. The design can rely on `prediction_requests` being a complete, countable log of requests per user (US-13). |

# 8 Technical architecture

## 8.1 Fixed stack

| Layer | Requirement |
|---|---|
| Hosting | **Vercel.** The web application and its server functions are deployed on Vercel. Preview deployments should be used for review before production release. |
| Database | **Neon Postgres.** All persistent data - users, profiles, countries, catalogue, selections, requests, and predictions - lives in Neon Postgres. Database credentials are held in environment configuration, never in source code. |
| Application | A single web application serving responsive pages for desktop and mobile browsers. Framework choice is a technical-design decision, provided it deploys cleanly to Vercel and connects to Neon Postgres. |
| Authentication | Email and password only for v1 (FR-01 to FR-04a), with hashed passwords and session handling appropriate to the chosen framework. |
| Environments | At minimum a production environment and a preview / development environment, each with its own Neon database or branch, so test readings, catalogue edits, and seed changes never touch production user data first (Section 4.5). |
| Settings | Operational settings that the operator tunes - the refusal rate (10% at launch), confidence weighting, the guest delay duration, and the active provider name - are stored as database settings or environment configuration, readable by the app at request time, and changeable without a code release. |

## 8.2 Proposed Neon Postgres schema

Logical schema for v1. Column types are indicative for technical design; primary keys are UUIDs unless the technical design records a reason to differ. Timestamps are stored in UTC and displayed in the profile timezone.

| Table | Key columns and purpose |
|---|---|
| `countries` | `code` PK (ISO 3166-1 alpha-2); `name`; `is_active`; `created_at`, `updated_at`. Reference data. A country is the join point between profiles and lottos. |
| `users` | `id` PK; `email` unique, not null; `password_hash` not null; `created_at`, `updated_at`, `last_login_at`. One row per account. No profile fields live here. |
| `profiles` | `user_id` PK / FK to `users` (one profile per user); `display_name`; `country_code` FK to `countries`, not null once onboarding is complete; `timezone` (IANA name, e.g. `Europe/London`); `date_of_birth` (date, information only, stored restricted - see FR-07 and NFR-02); `personal_touches` (JSONB, nullable - lucky numbers array, names / words list, significant dates with labels; see Section 6.7 - a separate table is an acceptable technical-design alternative, provided it is one-to-one with the profile, is never exposed on share pages, and deletes with the account); `disclaimer_acknowledged_at`; `onboarding_completed_at` (null until FR-05 is complete); `created_at`, `updated_at`. |

| Table | Key columns and purpose |
|---|---|
| `lottos` | `id` PK; `name`; `country_code` FK to `countries`, not null; `jurisdiction_note` nullable; `main_count`, `main_min`, `main_max`; `bonus_count`, `bonus_min`, `bonus_max`, `bonus_from_main_pool` boolean; `draw_days` (array / JSON list); `is_active` boolean; `created_at`, `updated_at`. Reference data, seeded (Section 4) and maintained by direct database access only (FR-16). |
| `user_lotto_selections` | `id` PK; `user_id` FK; `lotto_id` FK; `selected_at`. Unique constraint on (`user_id`, `lotto_id`) so a lotto cannot be saved twice to one profile. Deleting a row is how a user deselects a lotto - and how old-country selections are cleared on a country change (FR-13). |
| `prediction_requests` | `id` PK; `user_id` FK; `lotto_id` FK; `requested_at`; `status` (completed / refusal / error); `provider_name`, `provider_version`. One row per request, whatever the outcome (FR-23). This table is also the countable usage record a future tier design would rely on (TODO-04). |
| `predictions` | `id` PK; `request_id` FK unique; `user_id` FK; `lotto_id` FK; `outcome` (numbers / refusal); `main_numbers` (int array, null on refusal); `bonus_numbers` (int array, null on refusal / no bonus); `sentiment_message` (null on refusal); `confidence_level` 1-5 (null on refusal); `refusal_message` (null on numbers outcome); `created_at`. History (FR-23) reads from this table. The lotto name shown in history must survive later catalogue edits, so history rendering must not depend on the lotto row's current name alone. |
| `app_settings` | `key` PK; `value`; `updated_at`; `updated_by`. Holds the refusal rate (launch value `0.10`), confidence weighting, active provider selection, the personal-touch lucky-number inclusion flag (Section 6.7, off until D-06 is confirmed if the owner prefers), and `guest_delay` - the guest reviewing-sequence duration in seconds (proposed default 12, tunable range about 8-20; Section 6.5). `updated_by` supports the change-note discipline in Section 4.5. |

| Table | Key columns and purpose (v2.1 additions) |
|---|---|
| `share_links` | `id` PK; `prediction_id` FK to `predictions`, not null; `user_id` FK (owner, for management and cascade delete); `token` unique, long and randomly generated, unguessable (FR-42) - the token is the only key a public visitor needs; `is_revoked` boolean, default false; `created_at`, `revoked_at` nullable. The public share page reads the prediction **through** this table only, and only when `is_revoked = false`. An alternative design - a share token plus revoked flag directly on `predictions` - is acceptable if it enforces the same rules; the table design is preferred because it keeps public access separate from the owner's history data. |
| `guest_prediction_requests` | `id` PK; `country_code` FK to `countries`; `lotto_id` FK; `requested_at`; `status` / outcome (numbers / refusal / error); `provider_name`, `provider_version`. **Anonymous:** no user, email, name, or personal-touch data - ever. Guest number sets may be stored here for audit / abuse review or discarded after display, as technical design decides; they create no history, no profile, and no share link. Standard rate-limit counters (for example by network address, held only as long as the security control needs them) live outside this table's product data. |

**Referential rules.** Deleting a user deletes their profile (including personal touches and date of birth), selections, requests, predictions, and share links - this is what makes self-serve deletion (FR-25) a single cascading operation, and it is why a deleted account's share pages stop working (FR-41). Guest rows have no user to cascade from and contain no personal data. Deleting a lotto or country that has selections, profiles, or predictions is prohibited; retire with `is_active = false` instead (FR-17). A database constraint or check should prevent a selection being created where the lotto's `country_code` differs from the profile's `country_code`, as a backstop to the application-level filter in FR-09.

# 9 Page flow and sequences

## 9.1 Pages

| Page | Purpose |
|---|---|
| Landing / Login / Register | Explain the product with the disclaimer visible; register, log in, or start password reset. Email + password only (FR-04a). Also the entry point to **guest mode** for visitors without an account. |
| Guest flow - Kaspah asks | Public, no login. Step 1: Kaspah, in flowery language, asks "Where are you from?" (country list). Step 2: Kaspah asks which lotto - only that country's subset. Then the reviewing sequence (below). English only. |
| Guest reviewing / waiting | Public. Crystal-ball visual, rotating flowery progress lines, and the "for faster results, create an account and log in" message, for the configured `guest_delay` duration (Section 6.5). |
| Guest result (singular) | Public, request-scoped. One lotto, one outcome - numbers + stars + feeling, or a refusal - with disclaimer, followed by the second flowery create-account prompt (FR-36). No navigation to rows, history, or profile. |
| Public share page | Public, no login, reached only by a valid share link (FR-39). Shows the shared reading and the "Want your own numbers?" call to action. A revoked / deleted / invalid token shows only the plain unavailable message. |
| Onboarding - Profile details | Display name, date of birth, and timezone. Required before country and lotto steps (FR-06). |
| Onboarding - Country | "Where are you from?" A single required choice from the active `countries` list. **This step comes before any lotto is shown** (Section 9.2). |

## 9.1 Pages (continued)

| Page | Purpose |
|---|---|
| Onboarding - Lotto selection | Only the active lottos for the chosen country, with disclaimer acknowledgement. At least one selection required to finish (FR-09, FR-10). |
| Home - Rows (Ask Kaspah) | **The rows view (Section 6.4):** one row per saved lotto, in saved order, each row with its own request action and result state. Shows the disclaimer. No request counter, quota, or limit messaging exists on this page in v1, and no artificial delay is applied to logged-in requests. |
| Reading result (in a row) | Numbers result or refusal, exactly as defined in Sections 6.1 and 6.3, rendered inside the requesting row without disturbing other rows. After a refusal, a clear "ask again" action is present in that row. A Share action appears on a completed numbers result. |
| My Readings (history) | Past readings and refusals, newest first (FR-23), dated in the user's timezone. |
| My Profile | Edit display name, timezone, date of birth, and country (with the Section 4.2 warning flow); edit lotto selections (FR-11); account actions including logout. |
| My Profile - Personal Touches | Logged-in only. Add / edit / remove lucky numbers, names / words, and significant dates (FR-43), with the help text in Section 6.7. |
| My Readings - Share management | Part of History: for a completed numbers reading, generate / copy / revoke its share link (FR-38, FR-41) and see whether a link is active or revoked. |
| Settings - Delete Account | The self-serve deletion flow in Section 9.4 (FR-24, FR-25). |

## 9.2 Core sequence - onboarding, country first

1. Visitor registers with email + password and logs in.
2. The app routes them to onboarding: profile details first (display name, date of birth, timezone).
3. The app asks **"Where are you from?"** and saves the chosen country to the profile. No lotto content has been shown yet.
4. Only now does the app load the lotto subset: active lottos where `country_code` equals the profile country. The user selects at least one, acknowledges the disclaimer, and onboarding completes.
5. The user lands on Ask Kaspah. If their country's subset was empty, they land instead on the plain explanation in FR-12, with a complete profile and no reading function until a lotto for their country exists.

## 9.3 Core sequence - requesting a reading

1. User logs in. If onboarding is incomplete, the app routes them to the incomplete step; otherwise to Ask Kaspah.
2. On the rows view, the user requests a reading from the row of the lotto they want. There is no limit check in v1, because there is no limit (FR-18), and no artificial delay is applied.
3. The app creates a `prediction_requests` row.
4. The app calls the configured `PredictionProvider` (v1: `RandomPredictionProvider`) with the lotto's stored rules.
5. The provider returns a numbers outcome or a refusal outcome (Section 7.2), applying the 10% refusal draw first.
6. For a numbers outcome, the app validates the set against the lotto rules. A set that fails validation is not saved as a prediction or shown; the request is marked as an error.
7. The app saves the `predictions` row - numbers, flowery sentiment, and stars, or the refusal message - and marks the request completed or refusal.
8. The app shows the matching result **inside that row**, with the disclaimer, leaving every other row untouched. After a refusal, the user may immediately start again at step 2, from any row.
9. The reading is now visible in history.

## 9.4 Sequence - self-serve account deletion

1. User opens Settings / My Profile and chooses Delete Account.
2. The app explains, in plain system voice (not Kaspah's voice - this is a serious, irreversible action): the account, profile including date of birth, lotto selections, and full reading history will be deleted; deletion cannot be undone; the email address could later be used to register a fresh, empty account.
3. The user performs the deliberate confirmation action (FR-24) - a single tap on "Delete" is not enough.
4. The app deletes the user and, by cascade, their profile, selections, requests, and predictions (or anonymises any record it is required to retain, per FR-25), invalidates their sessions, and returns them to the landing page, logged out.
5. Attempting to log in with the deleted credentials fails exactly as it would for an unknown account.

## 9.5 Sequence - changing country

Follows Section 4.2 step for step: edit in My Profile, warning, explicit confirmation, old-country selections cleared, history untouched, re-selection for the new country required before the next reading.

## 9.6 Sequence - guest reading

1. Visitor, not logged in, chooses the guest entry point on the landing page.
2. Kaspah asks, in flowery in-character language: **"Where are you from?"** The guest picks a country from the supported list. The choice is held for this request only.
3. Kaspah then asks which lotto they want numbers for, offering **only** that country's active lottos. The guest picks one.
4. The app logs an anonymous guest request row, calls the configured provider with that lotto's rules, validates any numbers outcome exactly as for logged-in readings, and applies the same 10% refusal roll.
5. The **reviewing sequence runs for the configured guest delay**: crystal-ball visual, rotating flowery progress lines, and the faster-with-an-account message (FR-35). The result is not shown before the sequence completes.
6. The singular guest result page shows the one outcome, with disclaimer. Below it, the second flowery prompt invites the guest to create an account for faster processing later, with Create Account / Log In actions (FR-36).
7. If the guest leaves, nothing is retrievable: no history, no saved choice, no account was created.

## 9.7 Sequence - sharing and viewing a shared reading

1. Owner opens a completed numbers reading in its row or in History and chooses Share; the site creates a `share_links` row with a fresh unguessable token and shows the owner the link to copy.
2. Anyone opening the link reaches the public share page: the server looks the token up, checks it is not revoked and its owner account still exists, and renders **only** the FR-39 fields for that one reading.
3. The visitor reads the "Want your own numbers? Create an account and log in." call to action; choosing it goes to Register / Login, with no access to the owner's data.
4. Owner revokes the link from History at any time; from that moment the same URL renders only the plain unavailable message. Deleting the account has the same effect on all of its links, by cascade.

## 9.8 Sequence - personal touches in a reading

1. User saves or edits touches in My Profile - Personal Touches; they persist on the profile.
2. User requests a reading from a row. The provider receives the touches as part of the user context for that request only.
3. Sentiment text weaves in names / words (and a near significant date, where the library provides for it); lucky numbers are handled only under the D-06 proposed default and its configuration flag (Section 6.7).
4. The result, in its row and in History, may show the in-character disclosure if a lucky number was included. Share pages generated from that reading show **none** of the touches and no disclosure that names them (FR-40).

# 10 Non-functional requirements

| ID | Requirement |
|---|---|
| NFR-01 | **Security.** Passwords hashed with a current standard algorithm; all traffic over HTTPS; a user can access only their own profile, selections, and history; database credentials held in environment configuration only - which matters doubly because catalogue maintenance is by direct database access (Section 4.5). |
| NFR-02 | **Privacy.** Collect only what v1 needs: email, display name, country, timezone, date of birth, selections, readings, and - only if the user chooses to add them - personal touches. Personal touches are profile data with the same protections: never logged in plain text outside the profile, never shown to another user, never on a share page, and deleted with the account. Guest mode collects no personal data at all. Date of birth is sensitive: store it in the profile only, restrict who and what can read it, never log it, never display it outside the holder's own profile, and never use it for generation or marketing. No sale of personal data. Self-serve deletion (FR-24, FR-25) is the v1 mechanism for erasure. |
| NFR-03 | **Performance.** A logged-in reading result should return within 2 seconds under normal conditions, excluding a brief theatrical reveal animation, which must be skippable, and logged-in readings must never be given the guest artificial delay. The guest result is the deliberate exception: it is held for the configured `guest_delay` reviewing sequence (Section 6.5) even though the underlying result is ready sooner. Unlimited access (FR-18) makes this a real load consideration: a single enthusiastic user must not be able to degrade the service for others, and any protective rate-limiting at the infrastructure level must be invisible in normal use and must never be presented as a product limit. |
| NFR-04 | **Availability / reliability.** Rely on Vercel and Neon managed availability for v1; a provider or database failure shows a plain error and never a fabricated or partial reading. |
| NFR-05 | **Usability / accessibility.** Responsive on common phone and desktop sizes; numbers legible at a glance; confidence shown as flowery text *and* stars, so it never relies on symbols alone; rows remain understandable when read in order by a screen reader, with each row's lotto named before its numbers; the guest reviewing sequence must announce progress accessibly and must not trap focus; country and lotto lists usable on a small screen; body text and controls meet ordinary contrast and keyboard-use expectations. |
| NFR-06 | **Configurability.** Countries, the lotto catalogue, the refusal rate, confidence weighting, and message libraries are data or configuration, changeable without rewriting application logic. |
| NFR-07 | **Auditability.** Every reading records provider name and version, timestamp, lotto, and outcome, so any historical reading can be explained after launch. Catalogue and settings changes made by direct database access are covered by the operator change log (Section 4.5). |
| NFR-08 | **Time correctness.** All timestamps are stored in UTC and rendered in the profile timezone, so "today", reading dates, and next-draw wording are correct for a global user base - the same request must not show different dates to the same user on different screens. |

# 11 Communications - no notifications in v1

The app sends **no notifications in v1**: no email notifications, no push notifications, no draw reminders, no "Kaspah has a new reading" nudges, and no marketing email. Silence is the specified behaviour, not a missing feature. The only outbound email permitted is transactional account email that the user themselves triggered - password reset (FR-03) and, if the design includes one, a confirmation that an account deletion has completed, sent to the deleted account's address as a security record. Neither may carry marketing content, reading content, or an invitation to return. Adding any notification later is a scope change requiring its own decision, consent model, and opt-out design. Guest-mode account prompts and share-page calls to action are **on-page copy, not notifications**: the app sends nothing to a guest, and a share page contacts nobody.

# 12 Responsible use - required for v1

**Entertainment-only disclaimer (FR-24a).** Wording to the effect of: "The Great Kaspah is for entertainment only. Numbers are generated at random, no reading changes the odds of winning, and this app does not provide financial or gambling advice." It must appear at onboarding (with acknowledgement recorded on the profile), on the reading screen, and with every result.

**No guarantee of winning.** No screen, message, confidence phrase, star label, marketing copy, or poster-derived artwork used in the app may state or imply a guarantee, an improved chance, or a "system". Confidence describes Kaspah's in-character feeling only (Section 6.2) - globally, in every country the app is used in.

**Date of birth - information only, for everyone.** Date of birth is collected as profile information only (FR-07). It is **not** an age gate in v1: it does not block, warn, restrict, or change the experience of any user, it is not presented as age verification, and it plays no part in generation, personalisation, or marketing. The entertainment-only disclaimer and the responsible-use wording in this section are shown to **every** user - and, in guest mode and on share pages, to every visitor - regardless of any date of birth. Users are reminded in general terms that buying a lottery ticket is age-restricted in most places and that they must follow the rules where they live; the exact per-country legal wording and responsible-play links remain pending as D-02, reframed in v2.1 as a wording exercise only - it does not convert DOB into a gate unless a future, explicit decision says so.

**Tickets are bought elsewhere.** The user is solely responsible for purchasing their own ticket, through an authorised retailer or operator in their own country, if they choose to play the numbers at all. The app does not sell tickets, take money, hold funds, check results, or claim prizes in v1.

**Play-responsibly signpost.** The profile or footer should link to responsible-play resources. Because the product is global, a single link cannot serve everyone: links are maintained per country alongside the D-02 wording work, and until a country's link is confirmed, the generic entertainment-only wording above still applies in full.

**Spending.** V1 is free and contains no purchases, so the app itself creates no spending. Copy must nevertheless never encourage a user to spend more on tickets because of a reading, a confidence level, or a refusal - in particular, refusal copy must not suggest that asking again, or playing more, improves anything.

# 13 Acceptance criteria for v1

## 13.1 Accounts, profile, and country filtering

- A new user can register with email + password, complete onboarding, and reach Ask Kaspah without assistance. No social or magic-link option appears anywhere.
- Onboarding enforces the order in Section 9.2: profile details, then country, then disclaimer, then lottos. **No lotto is visible before a country is saved.**
- For test accounts in at least three different countries, the selection list contains **exactly** the active lottos for that account's country - no more, no fewer - verified against the `lottos` table.
- A user can save multiple lottos, and selections persist across logout and login and apply to the next reading request.
- Profile shows and allows editing of display name, country, timezone, and date of birth. Reading dates and "today" wording render in the profile timezone.
- Changing country shows the Section 4.2 warning before saving; on confirmation, old-country selections no longer appear and cannot be used, the user is required to re-select before reading, and pre-change history still displays with its original lotto names.
- A test account whose country has no active lottos receives the FR-12 explanation and is never offered another country's lottos.

## 13.2 Readings, confidence, refusal, and unlimited access

- Across a test run of readings for every active lotto in the default list, **100% of numbers results pass rule validation** (count, uniqueness, range, bonus, sorting).
- Every numbers result shows the flowery descriptive phrase prominently **and** the matching star rating, with phrase and stars always agreeing in level.
- Across a large test run (1,000 requests against a test lotto), the refusal share is consistent with the configured **10%** rate within ordinary statistical variation, and both outcomes appear correctly in history and in the database, logged through the same pipeline.
- After any refusal, the same user can immediately request again for the same lotto and receives a fresh outcome; no lockout, waiting message, or quota text appears.
- A scripted burst of repeated requests from one account is never blocked for volume in v1 - no daily cap, draw cap, paywall, or upgrade prompt appears at any request count.
- No screen presents confidence as a probability or claims improved odds; the disclaimer appears at all three required points.

### 13.3 Catalogue maintenance by direct database access

- Adding a new country row and a new lotto row (inserted inactive, validated per Section 4.5, then activated) makes that lotto selectable by a test account of that country **with no code release and no redeploy of application logic**.
- A lotto row with a missing rule or a non-existent country code cannot be activated into users' view - and if forced active in a test database, the operator checklist in Section 4.5 is shown to catch it before production.
- Retiring a lotto removes it from new selection while an existing user's history still renders that lotto's readings correctly.
- There is no admin URL, admin screen, or in-app catalogue editor in the shipped v1 build.
- A catalogue change performed for acceptance is recorded in the operator change log with date, person, and rows changed.

### 13.4 Account deletion

- A test user can delete their account from Settings without contacting anyone, and cannot complete deletion with a single tap - the deliberate confirmation step is required.
- After deletion: login with those credentials fails; the profile, selections, requests, and history rows for that user are gone from the database (or anonymised where retention is required, verifiably unlinked from the person's identity); and the same email can register a fresh account that shows no trace of the old data.
- The pre-deletion screen states, before confirmation, that history will be deleted and that deletion is irreversible.

### 13.5 Scope negatives - what must be absent

- No draw-result checking, match reporting, or winnings language anywhere in v1.
- No notification of any kind is sent during a full test cycle (registration, readings, refusals, country change) other than a user-triggered password-reset email.
- No payment, subscription, tier, or quota surface exists; the app is free for every user.
- No external prediction API is called in v1; all readings in the database record the random provider and its version.
- **Rows:** there is no primary-lotto shortcut, no combined "read all" result that merges rows, and no row for any lotto the user has not saved or for any other country.
- **Guest:** no guest history, guest rows, guest profile, guest personal touches, or guest share link exists; a guest cannot retrieve a past guest reading.
- **Language:** no language switcher, translated surface, or non-English Kaspah line ships in v1.
- **Date of birth:** no screen blocks, warns, or changes behaviour based on date of birth, and no screen calls DOB collection "verification".
- **Share privacy:** no share page contains an email, date of birth, timezone, personal touch, display name, or any account detail.
- **Delay:** no logged-in reading is artificially delayed; the guest delay never appears inside the logged-in app.
- The deployed app runs on Vercel against Neon Postgres, with no production credentials in source code.

### 13.6 Acceptance criteria - rows, guest mode, sharing, personal touches (v2.1)

- **Rows:** a test user with three saved lottos sees exactly three rows, in saved order, each labelled with its lotto and country. Requesting in the middle row changes only that row; the other two rows' states are byte-for-byte what they were. A refusal in one row leaves a numbers result in another row fully intact.
- **Guest flow order:** starting as a guest, no lotto is visible before a country is picked, and after picking a test country the list contains exactly that country's active lottos. Completing the flow creates no user, profile, selection, or history rows for that visitor.
- **Guest delay:** the guest result is not displayed until the reviewing sequence for the configured `guest_delay` has run; the sequence shows rotating flowery lines and the faster-with-an-account message throughout. The same reading requested while logged in returns without that delay. Changing the setting changes the guest wait without a code release.
- **Guest prompts:** both account prompts are present and tested - one during the wait (FR-35) and one after the result (FR-36) - each in flowery language with a working create-account / log-in action.
- **Sharing:** a share link generated from a numbers reading opens, logged out, a public page with exactly the FR-39 fields and the "Want your own numbers? Create an account and log in." call to action. Altering the token returns no reading. A refusal cannot be shared.
- **Share privacy and revocation:** inspecting a share page (including its page source) reveals no email, DOB, timezone, personal touch, display name, or account identifier. After the owner revokes the link, and separately after the owner deletes their account, the same URL shows only the unavailable message.
- **Personal touches:** touches save, edit, and delete on the profile; a reading's feeling text reflects a saved name / word; touches never appear on a share page for that same reading; and deleting the account deletes the touches. With the D-06 flag on, a generated set contains **at most one** lucky number, only if valid for that lotto, with the in-character disclosure and no-advantage clarification present; with the flag off, numbers are fully random.
- **DOB and language:** two test accounts differing only in date of birth receive identical access, wording, and readings behaviour. Every tested surface - UI, guest flow, share page, Kaspah lines - is in English, with no language switcher present.

# 14 Risks, assumptions, and open decisions

## 14.1 Risks

| Risk | Mitigation in this BRD |
|---|---|
| Users believe Kaspah improves their odds | Entertainment-only positioning throughout; confidence defined as character feeling - flowery words and stars, never probability (Sections 6.2, 12). |
| Global scope outruns legal wording: age limits and responsible-play rules differ per country | Country is the filtering key and DOB is collected as information only from day one (FR-06, FR-07); per-country wording is tracked as pending decision D-02 and must be confirmed country by country; the generic disclaimer applies everywhere in the meantime (Section 12). |
| Direct database editing introduces a bad lotto, a wrong country code, or a broken range - and it reaches users immediately | Named editors only, insert-inactive-then-validate, test-account check, backup or branch before bulk changes, operator change log (Section 4.5, FR-16). |
| Default lotto list is treated as final and its rules go stale | The list is labelled default and pending redefinition (D-01) in the data's own section (4.4); rules verified against operators before go-live and on redefinition. |
| Refusals frustrate users, or unlimited access is abused | Refusal rate is a modest, configurable 10% with an immediate ask-again path and no lockout (Section 6.3); unlimited access is a deliberate v1 product decision, with infrastructure-level protection kept invisible (NFR-03) and every request logged so real usage can inform a future tier design (TODO-04). |
| A country change silently strands a user's selections | Warning, explicit confirmation, and a forced re-selection step; history is never rewritten (Section 4.2, FR-13). |
| Date of birth is over-collected or misused | Information only, with no gate or secondary use (FR-07); restricted storage and display, deleted with the account (NFR-02, FR-25). The risk to control is scope creep - a future feature quietly turning DOB into a check - which FR-07 now expressly forbids without a new decision. |
| Future API creates scope creep into v1 | Explicitly out of scope; TODO-01 gates it behind a separate provider decision (Section 7.3). |
| A share link leaks private data, or is guessed / forwarded beyond the owner's intent | Share pages render only the FR-39 allow-list; tokens are long, random, and unguessable (FR-42); the owner can revoke at any time, and deletion kills all links (FR-41). Copy near the Share action must warn the owner, plainly, that anyone with the link can view that reading while it is active. |
| Guest mode is abused, or the deliberate delay simply annoys guests away | Anonymous logging plus standard infrastructure rate-limiting (FR-37) - a security control, not a quota. The delay is configurable (Section 6.5) so it can be shortened after launch without a rebuild if guest completion data says it is too long. |
| Personal touches are mistaken for a lucky system, or private words leak | Touches are framed as personalisation only, with the no-advantage clarification wherever a lucky number is disclosed (Section 6.7); touches never leave the owner's readings - not on share pages, not in guest mode, deleted with the account (FR-44). |
| Users expect their own language because the product is global | English-only is a stated v1 limit (FR-46) in the product description itself, so no user is promised a translation that does not exist; other languages are a recorded future consideration. |

## 14.2 Assumptions

- A-01: One account holds one profile; one email equals one user; one profile has exactly one country at a time.
- A-02: Users play - or choose not to play - the numbers entirely outside the app; the app's job ends at the reading.
- A-03: The default lotto list's published rules, as verified by the operator before go-live, are correct at launch; operators may change them later, which is a data update, not a defect.
- A-04: Timezones use IANA names supplied or confirmed by the user; the device pre-fill is a convenience, not a guarantee.
- A-05: A guest who is shown an account prompt understands it as an invitation, not a requirement - the guest result is delivered in full whether or not they ever register.
- A-06: A user who shares a link controls where they send it; the site's responsibility is the page the link opens, its privacy allow-list, and revocation - not the channels the user shares it through.

## 14.3 Open decisions - only what is genuinely pending

The v1 questions Q-01 to Q-05 are **closed** by the answers in the Section 1.3 changelog, and the **round-2 questions are likewise closed** by the answers in Section 1.4: rows presentation (no primary lotto), Kaspah-lines approach, DOB as information only, English only, guest mode, sharing, personal touches, and buddy sign-off are all firm requirements in v2.1. What remains open is:

| ID | Pending decision | Status / effect |
|---|---|---|
| D-01 | Final redefinition of the lotto list: which countries, which games per country, verified rules for each. | Pending. The default list in Section 4.4 ships meanwhile; redefinition is a data exercise when it lands. |
| D-02 | Per-country legal wording: exact responsible-play text and responsible-play links, country by country. **Reframed in v2.1:** date of birth is information only and no hard age gate is built in v1 - this item is about wording and links only, unless a future explicit decision changes that. | Pending. The generic disclaimer, shown to everyone regardless of DOB, ships in v1 (Section 12); countries should not be actively promoted before their wording is confirmed. |
| D-03 | Future API provider for TODO-01: provider, authentication, contract, cost, and fallback. | Pending - no provider chosen yet; the product owner's team is working on it. Nothing in v1 depends on it. |
| D-04 | Future free / paid tier design: what, if anything, a paid tier would include and whether free requests would ever be limited. | Pending and future only. V1 is unlimited and free; `prediction_requests` already provides the usage data a design would need. |

## 14.3 Open decisions (continued)

| ID | Pending decision | Status / effect |
|---|---|---|
| D-05 | Final Kaspah message library: the flowery sentiment lines per confidence level, the refusal lines, and the guest reviewing / post-result lines, in Kaspah's voice. | Pending final approval only - the **approach is confirmed**: the team drafts a starter library and **Paul's buddy, the face of Kaspah, gives final approval**. The example lines in Sections 6.2, 6.5, and 6.7 are the working draft for that approval. |
| D-06 | **Personal-touch number inclusion - exact mechanics.** Confirm, amend, or reject the proposed default in Section 6.7: at most one valid saved lucky number may be included in a generated main set, chosen at random, with in-character disclosure and the no-advantage clarification; bonus numbers never come from lucky numbers. | **New in v2.1 - proposal, not confirmed.** Owner: Paul, with buddy input. Sentiment weaving (names / words, significant dates) ships in v1 regardless; the number-inclusion behaviour is built behind a configuration flag so this decision can land after the rest of v2.1. |

**Confirmed - buddy sign-off.** Paul's buddy is on board and has loved the ideas in this document, so the persona and likeness direction - his face and character as The Great Kaspah - is **approved in principle** and is not an open decision. Housekeeping before go-live (informational, not a blocker): formal written consent for commercial use of his likeness should be on file, and a domain name for the site has **not yet been provided** - neither item blocks building to this BRD.

## 14.4 Success criteria for the MVP

V1 succeeds if a small group of real users - in more than one country - can complete the full loop without support: register, give their country, save their lottos, see them as rows, receive readings and occasional refusals row by row, share a reading with a link that leaks nothing private, ask again whenever they like, and return for a later draw, with zero invalid number sets and zero readings offered for the wrong country. Guest success is its own signal: a visitor can finish the guest flow, sit through Kaspah's reviewing with good humour, and understand - without being forced - why an account is faster. Sign-ups alone are not success; repeat reading requests across separate draw cycles, from users whose profiles, rows, and personal touches simply stayed correct, are the signal that the ritual works.